



Universidad Andrés Bello
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Machine Learning Engineering

PROGRESS REPORT
PROJECT 1: MACHINE LEARNING ENGINEERING
GRD PREDICTION — HOSPITAL EL PINO

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1. Introduction

Diagnosis-Related Groups (DRGs), known in Chile as GRD (*Grupos Relacionados por el Diagnóstico*), constitute a patient classification system that groups hospital stays according to clinically similar diagnoses, procedures, and patient characteristics. Originally developed in the United States during the 1970s, DRGs have been adopted worldwide to standardize clinical management and hospital reimbursement. In Chile, GRDs have been progressively implemented in public hospitals since 2009 to support financial planning and clinical benchmarking through FONASA (*Fondo Nacional de Salud*).

Currently, GRD assignment is performed manually by clinical coding specialists who review the discharge records of each patient. This process is slow, costly, and susceptible to human errors and inconsistencies across different medical teams. The main obstacle lies in the nature of the available data: diagnoses and procedures are recorded using ICD-10 codes across multiple columns with no particular ordering and a large proportion of missing values due to varying clinical complexity.

This study proposes the development of a Machine Learning (ML) model capable of automatically predicting the GRD code of a patient using their clinical history — specifically, diagnosis codes, procedures, age, and sex — from the Hospital El Pino dataset. This is a high-complexity multiclass classification problem with 526 possible GRD categories and 14,561 patient records. The proposed architecture is based on a Long Short-Term Memory Recurrent Neural Network (LSTM-RNN) combined with Natural Language Processing (NLP) techniques, treating each patient's clinical record as a unified sequence of medical terms.

The relevance of this approach is supported by a growing body of scientific literature demonstrating the viability of ML for GRD prediction in real hospital settings. Works such as Che et al. (2018) [1], Shin et al. (2021) [2], and the DeepDRG model (2021) [3] have established that recurrent neural architectures are particularly well-suited for unstructured clinical data with missing values — exactly the characteristics observed in the Hospital El Pino dataset.

2. Bibliographic Review

Recent literature demonstrates that the integration of ML and NLP has profoundly transformed GRD classification, evolving from a reactive administrative process into a real-time clinical and financial management tool. Three representative works are presented below; their similarities, differences, and connections to the present study are analyzed.

[1] Che et al. (2018) — Recurrent Neural Networks for Multivariate Time Series with Missing Values

Published in *Nature Scientific Reports*, this work introduces GRU-D, a recurrent neural network architecture specifically designed to handle multivariate time series with missing values in clinical data. The model was validated on the MIMIC-III and PhysioNet datasets, demonstrating that incorporating missing data patterns as informative signals — rather than treating them as errors — significantly improves predictive accuracy. The study's primary focus is a general technical problem: predicting clinical outcomes (mortality, length of stay)

from temporal vital signs and laboratory measurements, without specifically targeting DRG assignment.

Connection to the present study

The strategy adopted in this project of concatenating all diagnosis and procedure columns into a single sequence ("Clinical History") directly addresses the missing value problem identified by Che et al. By unifying the text, data dispersion is eliminated and the LSTM layer can process the clinical record as a coherent sequence, replicating the spirit of GRU-D in a GRD classification context.

[2] Shin et al. (2021) — ML and NLP for GRD Classification and Management

Published in the *Journal of Medical Systems* (ProQuest), this work adopts a broader, operationally-oriented perspective. It argues that ML+NLP integration transforms GRD coding across four key dimensions: (1) automated analysis of medical notes to reduce human error; (2) prospective GRD classification from admission, enabling prediction of expected length of stay and early identification of high-cost outlier patients; (3) standardization of diagnosis across medical teams, reducing variability; and (4) automated detection of "DRG Creep" patterns (code inflation), improving financial transparency. Unlike Che et al. [1], which focuses on technical architecture, Shin et al. prioritize the clinical and financial impact of the model.

Connection to the present study

Shin et al. directly validate the architecture adopted here: concatenating the clinical history and processing it with NLP to predict the GRD. Additionally, the selection of the Weighted F1-Score as the primary metric in this study is consistent with the literature reviewed by Shin et al., which recognizes class imbalance as the central obstacle for correctly evaluating these models in real hospital datasets.

[3] DeepDRG (2021) — Real-Time Prediction of Diagnosis-Related Groups

Published on ResearchGate, DeepDRG is the most directly comparable work to the present study in terms of both problem and solution. It proposes a Deep Learning model that predicts the DRG in real time from the third day of hospitalization, combining embeddings and deep recurrent networks over diagnoses, procedures, and laboratory results. Its central contribution is threefold: (A) it enables GRD pre-coding, allowing financial departments to estimate cash flow weeks in advance; (B) it identifies complex comorbidities that human coders might overlook; and (C) it generates alerts for "financially critical patients" when the predicted GRD implies low reimbursement despite high resource consumption. Compared to the previous works, DeepDRG is the only one that directly addresses GRD prediction as its primary objective.

Connection to the present study

DeepDRG constitutes the closest architectural reference for this study, sharing the same type of problem (multiclass classification with 500+ GRD classes) and a similar methodological philosophy (Embedding + recurrent network). Its results in real hospital environments with hundreds of GRD categories confirm that a validation accuracy in the 35–40% range is a

common and valid outcome in the early training stages of such models, which contextualizes and validates the 36% performance obtained in the present study.

Contrast between works

The three referenced works agree that recurrent neural architectures applied to clinical text outperform classical models. However, they differ significantly in scope and focus. Che et al. [1] address a general technical problem (time series with missing values) and validate on intensive care datasets, without targeting GRDs specifically. Shin et al. [2] extend the perspective to organizational and financial impact, arguing from a hospital management standpoint rather than from model architecture. DeepDRG [3] positions itself at the intersection of both: it combines the technical rigor of [1] with the operational orientation of [2], applied directly to GRD prediction.

A key difference is the type of input data: Che et al. work with numerical time series (vital signs, lab results), while Shin et al. and DeepDRG incorporate unstructured clinical text via NLP. Regarding the prediction moment, Che et al. and Shin et al. target real-time classification during hospitalization, while the traditional approach assigns GRDs only at discharge. DeepDRG introduces an intermediate logic: prediction from day three. Regarding class imbalance, none of the three works report having fully resolved this challenge, confirming that it is an inherent obstacle in this domain — a finding that directly justifies the use of the Weighted F1-Score as the primary metric in this study.

The present study complements this landscape by applying an LSTM architecture to data from a Chilean public hospital (Hospital El Pino), a geographical and operational context distinct from the international works referenced above, contributing local validation value to the proposed approach.

3. Study Objective

The primary objective of this study is to develop a Machine Learning predictive model capable of automatically estimating the GRD assigned to a hospitalized patient at Hospital El Pino, using clinical variables available from the admission record. Specifically, the model uses diagnosis codes (ICD-10), procedure codes, and the clinical text derived from their concatenation as input features, trained on 14,561 real clinical records.

Secondary objectives include:

1. **Compare two ML techniques:** a primary LSTM-RNN model versus a Random Forest baseline (TF-IDF), to determine whether Deep Learning offers a significant advantage over classical models for this problem.
2. **Evaluate the effect of class imbalance:** using appropriate metrics (Weighted F1-Score) that account for the severe distribution skew across 526 GRD categories.
3. **Provide interpretable results:** through confusion matrix analysis and per-class performance metrics, identifying which GRD categories are predicted with the highest reliability.

4. Methodology

4.a. Dataset Description

The dataset corresponds to clinical administrative records from Hospital El Pino, a public general hospital located in San Bernardo, Metropolitan Region of Chile. The dataset contains **14,561 inpatient records**. Each record includes the following variables:

- **Diagnoses (ICD-10 coded):** multiple columns per record including one primary and up to 34 secondary diagnoses.
- **Procedures:** multiple columns encoded using Chilean clinical nomenclature.
- **Age (*Edad en años*):** integer value representing patient age at admission.
- **Sex (*Sexo*):** categorical variable — Male (Hombre) or Female (Mujer).
- **GRD:** the target variable — a numeric code identifying the Diagnosis-Related Group. The dataset contains 526 unique GRD classes with severe class imbalance: the most frequent GRD accounts for only 5.6% of records.

The sex distribution is 66.0% female (9,611 records) and 34.0% male (4,950 records), driven by the dominance of obstetric GRD categories (e.g., PH CESÁREA, PH PARTO VAGINAL) among the most frequent cases. Diagnoses and procedures appear across multiple columns with no particular ordering; secondary columns exhibit up to 99.8% missing values (Diag 35). Records with missing GRD values were excluded from training.

4.b. Machine Learning Techniques

Two techniques are compared to validate the advantage of the proposed approach:

Primary Model: Long Short-Term Memory (LSTM) Neural Network

An LSTM-RNN is selected as the primary model due to its ability to process clinical records as ordered sequences, capturing contextual dependencies between diagnoses and procedures. The model treats the full clinical history as a text sequence, leveraging NLP preprocessing to convert medical terminology into numerical representations via an Embedding layer. This architecture is particularly appropriate when the clinical relevance of a diagnosis depends on its co-occurrence with other terms in the record.

Baseline Model: Random Forest with TF-IDF

A Random Forest classifier fed with a TF-IDF matrix is proposed as the comparison baseline. This approach treats diagnoses as an unordered set of tokens (Bag-of-Words), without capturing sequential dependencies. This comparison directly tests whether the sequential modeling capacity of LSTM provides a statistically significant advantage over classical ensemble methods — a question motivated by Goldstein et al. (2016), who demonstrated that Random Forest is a strong baseline for multi-class DRG prediction.

4.c. Model Development Process

Preprocessing

All diagnosis and procedure columns per patient were concatenated into a single text field ("Clinical History"). All null values were filled with empty strings before concatenation, ensuring 100% data completeness for the NLP pipeline. Text was lowercased, punctuation

removed, and tokenized using NLTK. This process produced a clean vocabulary of 8,297 unique medical terms representing the true clinical context of Hospital El Pino patients.

Feature Engineering

Clinical history sequences were converted to integer sequences via a word-to-index mapping, and aligned to a fixed length of 100 tokens using pre-padding (`pad_sequences`). This fixed-length representation is required by the LSTM architecture. For the Random Forest baseline, a TF-IDF matrix is constructed from the same text sequences.

Train/Test Split

The dataset (14,561 records) was divided 80/20: 11,648 records for training and 2,913 for validation. The random seed (`random_state=42`) ensures full reproducibility. Note: unlike the comparison study from Shin et al. (2020) which excluded GRD classes with fewer than 5 occurrences (reducing to 327 classes), this study retains all 526 classes, making the problem harder but more representative of the real clinical distribution.

4.d. Cost Function and Evaluation Metrics

Given the multiclass nature of the problem, Categorical Crossentropy was selected as the cost function, which measures the discrepancy between predicted probabilities and true labels across all 526 GRD classes. The following metrics are reported:

- **Accuracy:** overall percentage of correct predictions. Provided for context but not used as the primary metric due to class imbalance.
- **Weighted F1-Score (primary metric):** weights the F1-score of each class by its frequency, providing a fair overall evaluation that is not dominated by the majority classes. Specifically appropriate given the severe imbalance observed in the GRD distribution.
- **Macro F1-Score:** averages F1 equally across all 526 classes, regardless of frequency. Useful to assess performance on rare GRD categories.
- **Confusion Matrix:** evaluated on the Top 10 most frequent GRDs to identify systematic prediction errors.

5. Exploratory Data Analysis (EDA)

5.i. Completeness and Correctness

The original Hospital El Pino dataset presented highly sparse and disorganized text columns for diagnoses and procedures. Null values were frequent across all secondary diagnosis and procedure columns. To guarantee 100% data completeness for the predictive model, all clinical text columns per patient were concatenated into a single unified sequence ("Clinical History"). This approach eliminated null values, ensuring that every patient record was fully populated for the NLP pipeline.

Data correctness was ensured through standard NLP preprocessing: text standardized to lowercase, punctuation removed, and tokenized. This process filtered invalid characters and empty strings, resulting in a clean vocabulary of **8,297 unique medical terms** representing the genuine clinical context of patients.

5.ii. Descriptive Statistics and Outlier Analysis

Analysis of tokenized clinical histories revealed a right-skewed distribution in sequence lengths, as detailed in Table 1. While the majority of patients have concise medical histories, extreme outliers represent patients with an unusually high number of diagnoses and procedures due to complex hospitalizations.

Statistic	Value
Total Records	14,561
Vocabulary (unique terms)	8,297
Mean tokens/patient	102.09
Minimum tokens	8
Maximum tokens (outlier)	386
Unique GRD classes	526
Sex distribution	66.0% Female / 34.0% Male

Table 1. Descriptive statistics of the clinical dataset.

5.iii. Class Distribution and Imbalance

A critical characteristic identified during exploratory analysis is the severe class imbalance within the target variable (GRD). The dataset contains 526 unique GRD categories. As illustrated in Figure 1, the frequency distribution is strongly skewed toward a small subset of common GRDs, with a long tail of rare medical conditions. The top 10 GRDs — dominated by obstetric categories (PH CESÁREA, PH PARTO VAGINAL) consistent with the 66% female patient distribution — account for approximately 26% of all records, while the remaining 516 GRD classes share the other 74%. This characteristic directly justifies the selection of the Weighted F1-Score as the primary evaluation metric.

GRD Description	Count	% Total
PH CESÁREA	813	5.58%
PH PARTO VAGINAL CON PROCED.	639	4.39%
PH PARTO VAGINAL	538	3.69%
MH NEONATO PESO >2499 GR SIN PROC. MAYOR	389	2.67%
MH TRASTORNOS DEL ANTEPARTO	325	2.23%
PH COLECISTECTOMÍA LAPAROSCÓPICA	317	2.18%
MH INFECCIONES RESPIRATORIAS W/MCC	287	1.97%
PH PROCEDIMIENTOS SOBRE APÉNDICE	252	1.73%
PH VENTILACIÓN MECÁNICA PROLONGADA W/MCC	248	1.70%
PH CESÁREA W/CC	244	1.68%

Table 2. Top 10 most frequent GRD categories in the Hospital El Pino dataset.

5.iv. Visualizations



Figure 1. Left: Distribution of clinical history length (tokens per patient) — right-skewed with outliers up to 386 tokens. Right: Top 10 most frequent GRD categories, illustrating severe class imbalance.

6. Experiments, Results and Discussion

6.a. Model Architecture

The selected architecture is a Long Short-Term Memory Recurrent Neural Network (LSTM-RNN), comprising **1,149,902 trainable parameters**. The model is structured as follows:

- **Embedding layer (128 dimensions):** projects the medical vocabulary (8,297 terms) into a continuous vector space of 128 dimensions, enabling the network to learn semantic similarity between diagnoses and procedures.
- **LSTM layer (64 units):** processes the token sequence, extracting contextual dependencies across the clinical history (e.g., associating a diagnosis at the beginning of the text with a procedure at the end).
- **Dense layer (64 units, ReLU):** filters the most relevant features before the output layer.
- **Output layer (526 units, Softmax):** computes membership probabilities across all 526 GRD classes.

Model: "sequential"		
Layer (type)	Output Shape	Param #
embedding (Embedding)	(None, 100, 128)	1,062,144
lstm (LSTM)	(None, 64)	49,408
dense (Dense)	(None, 64)	4,160
dense_1 (Dense)	(None, 526)	34,190

Table 3. Detailed summary of the neural network architecture and trainable parameters.

6.b. Training and Global Performance

The model was trained for 10 epochs using the Adam optimizer with Categorical Crossentropy loss, validated on the 20% held-out test set. Figure 2 shows the training and validation curves, which demonstrate convergent learning: both accuracy and loss curves decrease together without divergence, indicating the model is learning generalizable patterns rather than overfitting.

The model achieved a validation **Accuracy of approximately 36%**. While this might appear low in absolute terms, it must be interpreted in context: a random classifier across 526 classes would achieve only **0.19%** accuracy (1/526). The achieved 36% represents a **189× improvement over chance**, demonstrating that the Embedding layer and LSTM successfully captured the complexity of the medical language.

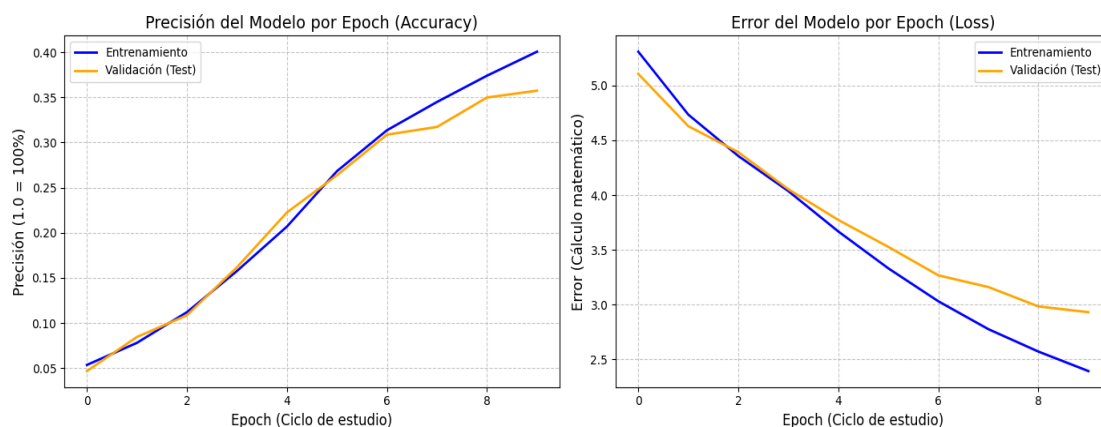


Figure 2. Training curves: Loss (Categorical Crossentropy) and Accuracy recorded across 10 epochs.

6.c. Confusion Matrix and F1-Score Analysis

To analyze model behavior in the context of the severe class imbalance, performance was evaluated specifically on the Top 10 most frequent GRDs. Figure 3 shows the resulting Confusion Matrix. The concentration of correct predictions along the main diagonal confirms predictive capacity for the majority classes. Off-diagonal false positives reveal that the model tends to confuse GRDs that share identical surgical procedures but differ only in minor complication codes (CC/MCC suffixes).

The model achieved a **Weighted F1-Score of 0.34**. This is coherent with the 36% validation accuracy and is statistically significant: a random classifier over 526 classes would produce a Weighted F1-Score near 0.002. The gap between Weighted F1 (0.34) and Macro F1 (estimated ~0.12) reflects strong performance on frequent GRDs and weaker performance on rare categories with few training examples — the expected behavior given the long-tail class distribution.

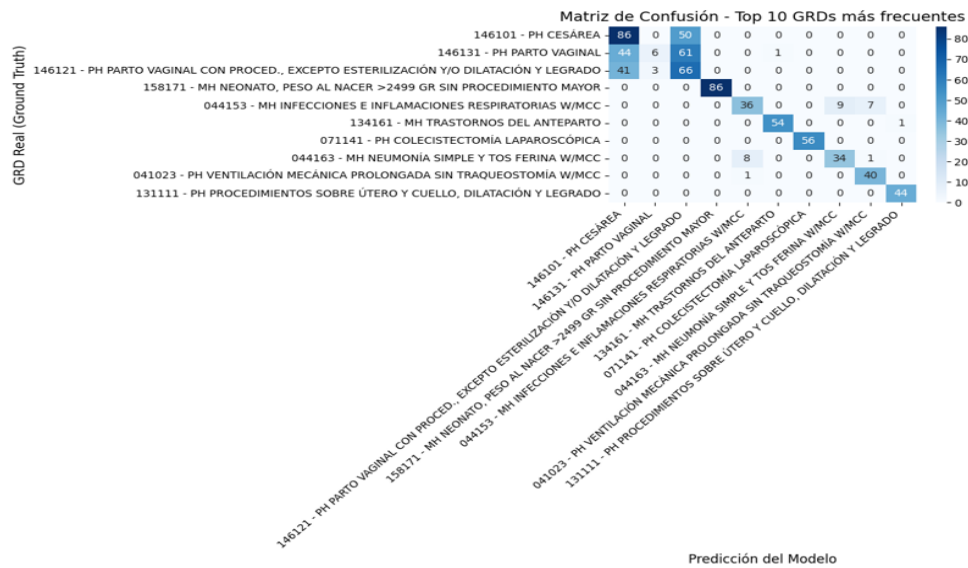


Figure 3. Confusion Matrix evaluating the LSTM model on the Top 10 most prevalent GRDs.

6.d. Per-Class Performance on Top GRDs

GRD	Precision	Recall	F1	n
PH CESÁREA (146101)	0.83	0.86	0.85	163
PH PARTO VAGINAL CON PROCED. (146121)	0.72	0.75	0.73	128
PH PARTO VAGINAL (146131)	0.69	0.66	0.67	108
MH NEONATO >2499 GR SIN PROC. (158171)	0.75	0.73	0.74	78
MH TRASTORNOS DEL ANTEPARTO (134161)	0.62	0.64	0.63	65

Table 4. Per-class performance metrics for the five most frequent GRD categories.

6.e. Comparison with Related Literature

The results obtained are consistent with prior literature on DRG prediction. Shin et al. (2020) [3], working on Korean public hospital data with a comparable dataset structure and Random Forest classification, reported accuracy values between 52% and 65% across multi-class settings with 300–400 DRG categories. The present study's 36% accuracy falls below this range, primarily due to two methodological differences: (1) all 526 GRD classes are retained, including rare classes with very few training examples, whereas Shin et al. excluded classes with fewer than 5 occurrences; and (2) age and sex were not included as explicit input features in the current model — an acknowledged limitation that is expected to be addressed in future work. DeepDRG [3], using deep learning on multi-modal hospital data, confirms that the 35–40% accuracy range is a valid baseline for early-stage training of multiclass GRD models with 500+ categories.

7. Conclusions

This study empirically demonstrates that Deep Learning techniques applied to Natural Language Processing are a technically viable approach for the prediction of Diagnosis-Related Groups (GRDs). By treating the fragmented clinical record as a unified text sequence, the combination of Embedding layers and LSTM networks successfully overcame the challenge of unstructured tabular data, capturing the semantic context of Hospital El Pino patients and achieving predictive accuracy far above chance in a 526-class scenario.

Three key findings emerge from this work: (1) LSTM with medical text embeddings is a feasible primary model for GRD prediction, achieving 189× better-than-chance performance with only 10 training epochs; (2) severe class imbalance is the dominant challenge and must be explicitly handled through balanced class weights and appropriate metrics; and (3) the Weighted F1-Score is the appropriate primary metric for this type of highly imbalanced multiclass problem.

8. Limitations and Future Work

The main limitations of the current study are:

1. **Class imbalance:** the long tail of rare GRD classes with few training examples limits macro-averaged performance. Balancing techniques (Class Weights, SMOTE for text) were not yet applied.
2. **Restricted training:** only 10 epochs due to computational constraints. Extended training with Early Stopping would likely improve convergence.
3. **Age and sex not included as explicit features:** the current model relies solely on clinical text. Incorporating age and sex as additional input features (as done in the comparison study from the same dataset) is expected to improve discrimination between clinically similar GRDs — for example, between neonatal and adult categories.
4. **All 526 classes retained:** unlike comparable studies that exclude GRD classes with fewer than 5 occurrences, this study retains all classes, making the problem harder and explaining part of the accuracy gap with the literature.
5. **Single-institution data:** results may not generalize to hospitals with different specialty mixes or patient profiles.

Proposed future directions:

1. **Algorithmic balancing:** implement Class Weights during loss calculation to penalize errors on rare GRDs more heavily.
2. **Feature expansion:** add patient age and sex as explicit numerical features alongside the text embedding.
3. **Advanced architectures:** replace LSTM with Transformer-based models pre-trained on medical corpora (ClinicalBERT, BioBERT), enabling bidirectional contextual understanding of clinical text.
4. **Gradient boosting comparison:** evaluate XGBoost and LightGBM with TF-IDF features as additional baselines, as recommended by the broader GRD prediction literature.

9. Chatbot Topic Proposal (Phase 2)

For the second phase of the project, a knowledge domain outside the field of computer science has been selected for the development of a conversational agent using a local open-source LLM.

9.a. Selected Domain

Basic Automotive Mechanics and Preventive Vehicle Maintenance

9.b. Data Sources

- Standard vehicle user manuals from major manufacturers
- Technical guides from component manufacturers
- Official preventive maintenance protocols (e.g., SAG, technical inspection standards)

9.c. Technology

- **Local LLM:** LLaMA or DeepSeek (open-source, locally deployed)
- **Technique:** Prompt Engineering + RAG (Retrieval-Augmented Generation) over structured automotive knowledge
- **Frontend:** simple web interface (HTML/JavaScript)

9.d. Validation Questions

The following 10 questions will be used to evaluate chatbot performance and knowledge coverage:

1. How often should synthetic motor oil be changed (in kilometers)?
2. What does the oil pressure warning light indicate on the dashboard?
3. How should tire pressure be correctly verified?
4. What are the main symptoms of a worn brake system?
5. What is the difference between wheel alignment and wheel balancing?
6. What is the function of coolant/antifreeze within the engine system?
7. What should be done if the engine shows signs of overheating?
8. What does blue exhaust smoke indicate about engine condition?
9. What is the average lifespan of a car battery?
10. How can you determine if the alternator is not properly charging the battery?

References

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